



**COMSATS University Islamabad, Sahiwal
Campus**

Project Proposal

(SCOPE DOCUMENT)

for

Pilot Review

Version 1.0

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SCOPE DOCUMENT REVISION HISTORY

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Supervisor Signature: _____

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Contents

COMSATS University Islamabad, Sahiwal Campus	1
PilotReview	1
<i>By</i>	1
SCOPE DOCUMENT REVISION HISTORY	1
Abstract	Error! Bookmark not defined.
1. Introduction.....	4
2. Problem Statement.....	4
3. Problem Solution for Proposed System	5
4. Related System Analysis/Literature Review	5
5. Advantages/Benefits of Proposed System	6
6. Scope.....	6
7. Modules.....	7
7.1 Module 1: Authentication and User Management	7
7.2 Module 2: Business and Google Integration.....	7
7.3 Module 3: Review Fetching and Notifications	7
7.4 Module 4: AI Response Generation.....	8
7.5 Module 5: Response Management and Posting	8
7.6 Module 6: Analytics Dashboard	8
8. System Limitations/Constraints	9
9. Software Process Methodology	9
10. Tools and Technologies.....	9
11. Project Stakeholders and Roles	10
12. Team Members Individual Tasks/Work Division.....	11
13. Data Gathering Approach.....	11
14. Concepts.....	11
15. Gantt chart	13
16. Mockups.....	14
17. Conclusion	17
18. References.....	17
19. Plagiarism Report	18

Project Category: (Select all the major domains of proposed project)

- ☐ **A**-Desktop Application/Information System ☒ **B**-Web Application/Web Application based Information System
- ☒ **C**-Problem Solving and Artificial Intelligence Application ☐ **D**-Simulation and Modeling ☐ **E**-Smartphone Game
- ☐ **G**-Networks ☐ **H**-Image Processing ☐ Other (specify category) _____

Abstract

ReviewPilot is an AI-powered web application that helps local businesses such as restaurants, salons, clinics, and gyms manage and answer their Google reviews. The system connects to a business owner's Google Business Profile through OAuth 2.0, automatically fetches new reviews, and uses the Google Gemini large language model to draft a personalized, context-aware reply for each one. The owner reviews every draft on a dashboard and can approve, edit, regenerate, or skip it before it is posted back to Google.

In addition to response automation, ReviewPilot provides analytics on review volume, sentiment, rating distribution, and response rate, and sends email notifications when new reviews arrive. The application is built with the MERN stack (MongoDB, Express.js, React, Node.js) and aims to give small business owners an affordable way to protect their online reputation while saving the time normally spent writing replies.

1. Introduction

This document is the project proposal (scope document) for ReviewPilot, an AI-powered web application that helps local businesses respond to their Google reviews. It defines the problem, the proposed solution, the scope, the modules, the tools, and the plan of the project, and it forms the baseline for the SRS and design documents that follow. Online reviews strongly influence how customers choose restaurants, salons, clinics, gyms, and other local businesses. Google Maps and Search are usually the first place customers look, so the star rating and the owner's replies shape the reputation of the business. Replying to reviews is widely regarded as good practice because it shows customers that the business listens, and it is generally considered helpful for local visibility. However, most small business owners lack the time, staff, or writing confidence to reply consistently, so many reviews stay unanswered for weeks. Generic template replies tend to sound robotic and can frustrate customers, especially after a negative review. ReviewPilot addresses this by connecting to the owner's Google Business Profile, fetching new reviews automatically, and drafting a personalized reply for each review with a large language model (Google Gemini). The owner stays in control and can approve, edit, regenerate, or skip every draft before it is posted. The system also provides analytics on ratings, sentiment, and response rate. It is a full-stack web application built with the MERN stack and is developed as a project for BS Software Engineering at COMSATS University Islamabad, Sahiwal Campus. The remainder of this document follows the structure of the university's scope document template.

2. Problem Statement

Local businesses receive customer reviews on Google every week, but most owners do not have a dependable way to answer them. Writing a thoughtful reply to each review takes time that owners would rather spend running the business. Many owners are also unsure how to word a reply, particularly to a negative or mixed review. As a result, reviews stay unanswered, replies are inconsistent in tone, and unhappy customers feel ignored. Owners who do reply often paste the same generic message, which reads as impersonal. Owners must also log in to Google Business Profile repeatedly to notice new reviews, so timely responses are missed. There is no simple view that shows how many reviews are pending, how sentiment is changing, or how the response rate is trending. Enterprise reputation-management platforms exist, but they are often priced and designed for larger multi-location companies rather than small local owners. Google's own tools allow manual replies only and offer no AI assistance. ReviewPilot is being developed to fill this gap with a lightweight, affordable tool that keeps the owner in control. Although review-management products already exist, re-implementing the core idea as a student project gives hands-on experience with real integration work that is rarely covered in lectures. Through this project the team expects to learn full-stack development with the MERN stack, including REST API design and MongoDB data modeling. The team also expects to learn secure OAuth 2.0 integration with Google APIs and safe storage of access tokens. Working with a large language model will teach prompt engineering, output control, and handling of API cost and rate limits. Finally, the project builds skills in requirements engineering, scheduled background jobs, and analytics dashboards.

3. Problem Solution for Proposed System

ReviewPilot solves these problems by automating everything except the owner's final decision. The owner registers and links a Google Business Profile through Google OAuth 2.0, so no Google password is ever shared. The owner then selects which business locations to monitor. A scheduled job checks Google for new reviews every 30 minutes and stores them without duplicates. For each new review, the system classifies the sentiment as positive, negative, neutral, or mixed. It then builds a prompt from the business name, category, chosen tone, reviewer name, rating, review text, keywords, and sign-off name. The prompt is sent to the Google Gemini API, which returns a natural, personalized reply of two to four sentences. Replies to positive reviews are grateful, replies to negative reviews are apologetic, and neutral reviews receive a balanced reply, and the system never offers discounts or free items. The owner receives an email notification containing the review and the drafted reply, with a link to the dashboard. From the dashboard the owner can approve a reply in one click, edit it, regenerate it, or skip it. Approved replies are posted to Google through the Business Profile API and marked as posted. Per-business settings let the owner control tone, language, keywords, sign-off name, and an optional auto-approve rule for positive reviews. An analytics dashboard shows total reviews, pending responses, response rate, average rating, review volume, sentiment breakdown, and rating distribution. Together, these features reduce the time spent on replies and keep the tone consistent while the owner keeps control.

4. Related System Analysis/Literature Review

Several existing products help businesses manage online reviews. Four representative systems are discussed below.

Birdeye. Birdeye is a commercial reputation-management platform that gathers reviews from many sites and offers messaging and review-request tools. It is a broad suite, so it can be complex and costly for a single small business that only needs Google review replies. Its plans are generally aimed at growing and multi-location companies.

Podium. Podium combines review collection, web chat, and customer messaging in one paid platform. Its focus is customer communication rather than focused, owner-approved reply drafting. Small owners may pay for many features they will not use.

ReviewTrackers. ReviewTrackers monitors reviews across multiple platforms and provides reporting and alerts for larger organizations. It is built mainly for enterprises and multi-location brands. Its emphasis on monitoring and reporting makes it heavier than a small owner needs.

Google Business Profile (built-in tools). Google lets verified owners read and reply to reviews directly. Replies must be written manually, and there is no drafting help, sentiment view, or consolidated pending list. Owners must also remember to check for new reviews themselves.

The related system analysis shows that existing tools are either broad, expensive suites or manual tools with no assistance. ReviewPilot contributes a focused, low-cost, AI-assisted workflow centered on Google reviews, with the owner approving each reply.

Table 1 Related System Analysis with proposed project solution

Application Name	Weakness	Proposed Project Solution
Birdeye	Broad and complex suite; pricing tends to target larger or multi-location businesses.	ReviewPilot focuses only on Google reviews with a simple dashboard, built on free-tier services to keep cost low.
Podium	Emphasis on messaging and chat; reply drafting is not the core feature; paid bundle of features many owners do not need.	ReviewPilot concentrates on AI-drafted, owner-approved replies with tone, keyword, and sign-off control.
ReviewTrackers	Enterprise-oriented monitoring and reporting; heavier than small owners need.	ReviewPilot offers lightweight analytics (sentiment, rating distribution, response rate) tailored to single or few locations.
Google Business Profile (built-in)	Manual replies only; no AI drafting, no sentiment view, no consolidated pending list.	ReviewPilot fetches reviews automatically, drafts replies with Gemini, classifies sentiment, and shows all pending reviews in one place.

5. Advantages/Benefits of Proposed System

- **Saves time:** AI drafts a reply for every review, so owners only need to check and approve.
- **Personalized replies:** each reply uses the reviewer's name, rating, review text, and the business's own keywords and sign-off.
- **Consistent, professional tone:** owners choose a professional, friendly, casual, or formal tone that is applied to every reply.
- **Owner stays in control:** nothing is posted without approval unless the owner enables the optional auto-approve rule.
- **Faster response times:** reviews are fetched every 30 minutes and the owner is notified by email.
- **Actionable insights:** the dashboard shows sentiment, rating distribution, review volume, and response rate.
- **Low cost:** the system is designed to run on free tiers of hosting, database, and AI services, making it accessible to small businesses.

6. Scope

The scope of ReviewPilot covers a responsive web application for local business owners together with a small administrator area. Users can register, log in, and manage a profile using JWT-based sessions. Users can link a Google Business Profile through OAuth 2.0 and import one or more business locations. For each business, users configure the response tone, language, sign-off name, and SEO keywords. The system fetches new Google reviews automatically every 30 minutes and also lets the owner trigger a manual fetch. Every new

review with text is classified for sentiment and receives an AI-drafted response from Google Gemini. Owners can approve, edit, regenerate, or skip each draft, and approved replies are posted to Google. The status of every review is tracked as pending, ai_generated, approved, edited, posted, or skipped, and a full response history is kept. An analytics dashboard presents key metrics and three charts. Email notifications alert owners about new reviews, and owners can opt out. An administrator can view and manage user accounts and monitor platform usage. The following items are outside the scope of this version: reviews from other platforms such as Yelp or Facebook, payment processing and subscription billing, AI responses in languages other than English, and a native mobile application, since the interface is responsive web only. Real-time WebSocket updates are a possible future enhancement and are not part of the first release. The system depends on the availability of the Google Business Profile API and the Gemini API. All modules will be developed, tested, and deployed using free-tier cloud services.

7. Modules

ReviewPilot is divided into six modules. The special feature of the project is the AI Response Generation module, which drafts sentiment-aware replies using a large language model, together with the owner-approval workflow around it.

7.1 Module 1: Authentication and User Management

This module controls who can access the system and what they can do. A new user registers with a name, email, and password, and the system checks that the email is unique. Passwords are hashed with bcrypt before they are stored. On login the system returns a JWT that is validated on every protected request and expires after seven days. Users can view and update their profile information. The module supports two roles: business owner and administrator. Administrators can list users and activate or deactivate accounts. Access rules ensure that owners can only see their own businesses and reviews.

7.2 Module 2: Business and Google Integration

This module connects ReviewPilot to the owner's Google Business Profile and manages business locations. The owner clicks a connect button and is redirected to Google's consent screen using OAuth 2.0. After the owner grants permission, the system exchanges the authorization code for access and refresh tokens and stores them encrypted. Expired access tokens are refreshed automatically with the stored refresh token. The system lists the owner's locations and lets the owner choose which ones to monitor. For each business, the owner sets the response tone, language, sign-off name, SEO keywords, and the optional auto-approve rule for four- and five-star reviews. Owners can also deactivate a business to stop fetching reviews, or remove it entirely.

7.3 Module 3: Review Fetching and Notifications

This module keeps ReviewPilot in sync with Google and informs the owner. A scheduled job runs every 30 minutes and requests new reviews for each active business from the Google Business Profile API. Each review is stored with the reviewer name, rating, comment, date,

and photo URL, and duplicates are prevented using Google's review identifier. Reviews that already have a reply on Google are detected and skipped. If Google rate limits a request, the module retries with exponential backoff and logs failures without crashing the application. Owners can also trigger a manual fetch for one business. When a new review is detected, the module sends an email containing the reviewer, rating, review text, the AI-drafted reply, and a link to the dashboard. Duplicate notifications are never sent, and owners can opt out of emails.

7.4 Module 4: AI Response Generation

This module is the core of the project. For each new review with text, it first classifies the sentiment as positive, negative, neutral, or mixed. It then builds a prompt that includes the business name and category, the tone setting, the reviewer name, the rating, the review text, the SEO keywords, and the sign-off name. The prompt is sent to the Google Gemini API, which returns a reply of two to four sentences that sounds natural and human. The reply style follows the sentiment: grateful for positive reviews, apologetic for negative reviews, and balanced for neutral ones. The module never offers discounts or free items in a reply. Failed requests are retried up to three times with exponential backoff. Owners can regenerate a reply if they are not satisfied, and token usage is recorded for every generated reply.

7.5 Module 5: Response Management and Posting

This module gives the owner control over what is finally published. Pending reviews are shown in a list together with their AI-drafted replies. The owner can approve a reply with one click, edit it before approving, regenerate it, or skip the review. Each review carries a status of pending, ai_generated, approved, edited, posted, or skipped. When a reply is approved, the module posts it to Google through the Business Profile API and marks it as posted only after Google confirms success. If the business has auto-approve enabled, replies to four- and five-star reviews follow the same posting path without manual approval. A history view lists all approved, edited, posted, and skipped responses. The review workflow is designed so that an owner needs no more than two clicks to approve a reply.

7.6 Module 6: Analytics Dashboard

This module turns stored review data into insights for the owner. The dashboard shows four summary cards: total reviews, pending responses, response rate, and average rating. The response rate is calculated as responded reviews divided by total reviews, multiplied by 100. A line chart shows review volume over the last 30 days. A donut chart shows the sentiment breakdown into positive, negative, neutral, and mixed reviews. A bar chart shows the rating distribution from one to five stars. Owners with several businesses can filter the data by business and change the date range. The statistics are computed with MongoDB aggregation queries and refresh whenever new reviews are fetched or responses are posted.

8. System Limitations/Constraints

- **Google API access and quotas:** the Google Business Profile API requires business verification and approval, and it enforces daily quota limits that may delay access or slow fetching.
- **AI service limits and cost:** Gemini API usage is limited by free-tier quotas and per-token costs, so prompts must be kept short and repeated calls minimized.
- **Platform and language:** the first version supports only Google reviews and English responses, and it is a responsive web application without a native mobile app.
- **Zero budget:** as a student project, the system must use free tiers of hosting, database, and email services, which limits performance and concurrent users.

9. Software Process Methodology

ReviewPilot will be developed with an Agile, iterative process combined with the Object-Oriented methodology. Requirements are captured as use cases and modeled with UML, and the system is organized into modules, models, and services that map to real entities such as users, businesses, and reviews. Development is divided into short iterations, each delivering a working module that is tested before the next begins. This methodology was chosen because the project depends on external APIs whose behavior is only fully understood during implementation, so early feedback and the ability to adjust are important. It also suits the team's experience with modular, component-based web development.

10. Tools and Technologies

The following hardware/software tools and technologies will be used in the implementation of the project.

Table 2 Tools and Technologies for Proposed Project

Tools	Version	Rationale
Visual Studio Code	Latest stable	IDE for frontend and backend code
MongoDB Atlas / Compass	MongoDB 6.0+	Cloud DBMS and database GUI
Postman	Latest stable	API testing
Git and GitHub	Latest stable	Version control and collaboration
Vercel / Render (or Railway)	Free tier	Hosting for frontend and backend
Figma	Web version	UI/UX design and mockups
MS Word	2021 / Microsoft 365	Documentation
MS Project	2021 / Microsoft 365	Gantt chart and scheduling
MS PowerPoint	2021 / Microsoft 365	Presentation

Technology	Version	Rationale
JavaScript (Node.js)	Node.js 18+	Backend runtime and programming language
Express.js	4+	Backend web framework and REST API
React with Vite	React 18+, Vite 5+	Frontend framework and build tool
Tailwind CSS and shadcn/ui	3+ / Latest	Responsive styling and UI components
Recharts	2+	Analytics charts
MongoDB with Mongoose	MongoDB 6+, Mongoose 7+	Database and object modeling
JWT and bcryptjs	Latest	Authentication and password hashing
Google Business Profile API and OAuth 2.0 (googleapis)	Latest	Fetch reviews, post replies, authorize access
Google Gemini API (@google/generative-ai)	Latest	AI response generation
Nodemailer	6+	Email notifications over SMTP
node-cron	3+	Scheduled review fetching
Jest and Supertest	Latest	Automated testing
HTML5 / CSS3	5 / 3	Web markup and styling

11. Project Stakeholders and Roles

Table 3 Project Stakeholders for Proposed Project

Project Sponsor	COMSATS University, Islamabad
Stakeholder	Manal Hussain (SP24-BSE-056): Student, developer of Modules 1 to 3. [Student Name 2] (SP24-BSE-xxx): Student, developer of Modules 4 to 6. Project Supervisor: [Supervisor Name] Final Year Project Committee: Evaluation of project Client: Freelance web/software developer (requirements and feedback) End users: Local business owners; System administrator

12. Team Members Individual Tasks/Work Division

Table 4 Team Member Work Division for Proposed Project

Student Name	Student Registration Number	Responsibility/ Modules
Manal Hussain	SP24-BSE-056	Module 1 to Module 3: Authentication and User Management, Business and Google Integration, Review Fetching and Notifications. Backend APIs, OAuth 2.0 integration, and scheduled jobs.
[Student Name 2]	SP24-BSE-xxx	Module 4 to Module 6: AI Response Generation, Response Management and Posting, Analytics Dashboard. Gemini integration, approval workflow, and charts.

Both team members share requirements documentation, UI design, testing, and deployment.

13. Data Gathering Approach

Requirements will be gathered through short interviews with owners of local businesses such as restaurants and salons to understand how they currently handle Google reviews. A brief questionnaire will collect information about how often reviews arrive, how long replies take, and which features owners value most. The team will also study the review-management workflow in Google Business Profile and analyze existing tools to identify gaps. The official Google Business Profile and Gemini API documentation will be used to confirm technical capabilities and limits. Feedback from the client and the supervisor will be collected after each iteration.

14. Concepts

Concept-1: OAuth 2.0 Authorization. OAuth 2.0 lets a user grant an application limited access to their Google account without sharing a password. In ReviewPilot it is used to read reviews and post replies on the owner's behalf. The project involves handling consent, authorization codes, access tokens, and refresh tokens securely.

Concept-2: Large Language Models and Prompt Engineering. A large language model generates text from an instruction called a prompt. ReviewPilot designs prompts that combine business context, tone, and review data so replies are natural and safe. The team will learn to control length, style, and content of the output.

Concept-3: Sentiment Analysis. Sentiment analysis identifies whether a piece of text is positive, negative, neutral, or mixed. ReviewPilot uses it to choose an appropriate reply style and to build the sentiment analytics chart.

Concept-4: REST APIs and Rate Limiting. REST APIs allow separate systems to exchange data over HTTPS. The project consumes Google and Gemini APIs and exposes its

own API to the React frontend. It also applies exponential backoff, which increases waiting time between retries when a service limits requests.

Concept-5: Scheduled Background Jobs. Cron jobs run tasks automatically at fixed intervals. ReviewPilot uses one to fetch new reviews every 30 minutes. The team will learn to make such jobs reliable, recoverable, and safe to run in parallel batches.

15. Gantt chart

The Gantt chart below gives the estimated start and end dates of all modules/tasks for each team member, together with task dependencies (arrows show which task must finish before the next can start).

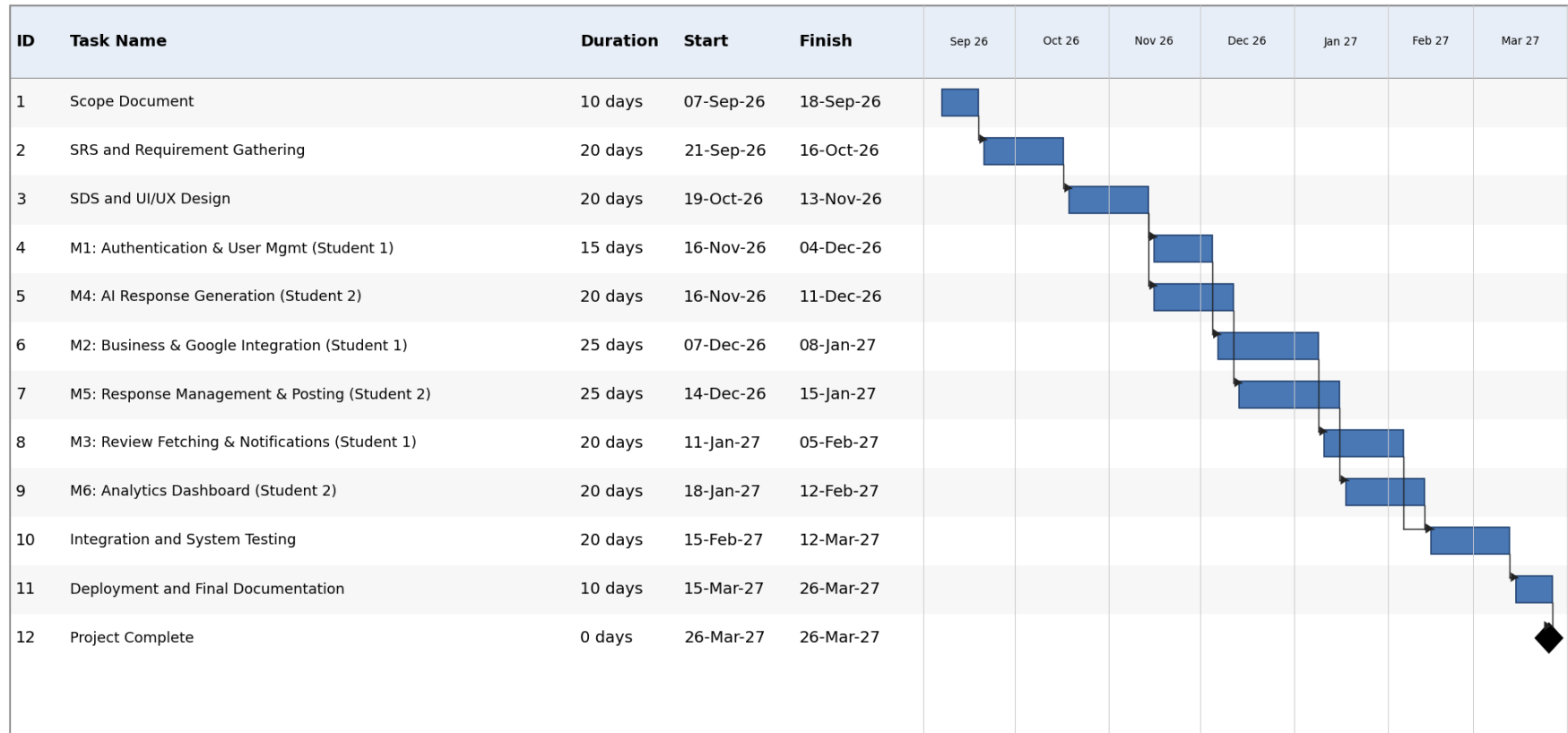


Figure 1 Gantt chart of ReviewPilot

16. Mockups

The mockups below show the major modules described in the scope section. Login, sign-up, and similar standard pages are intentionally not included.

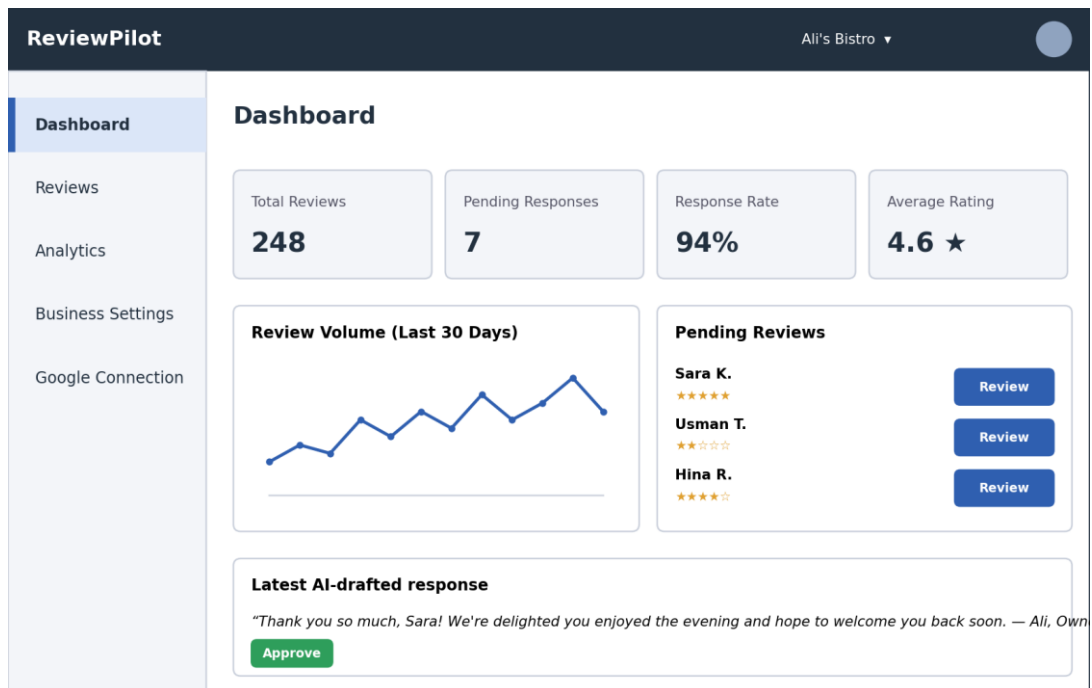


Figure 2 Dashboard mockup

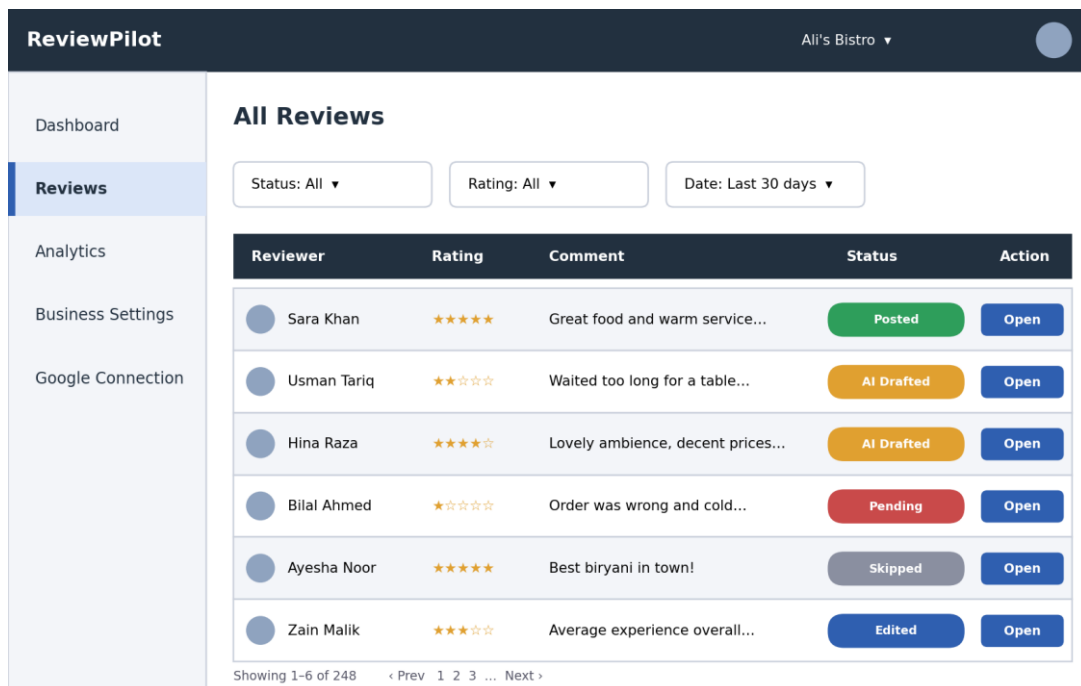


Figure 3 Reviews list mockup

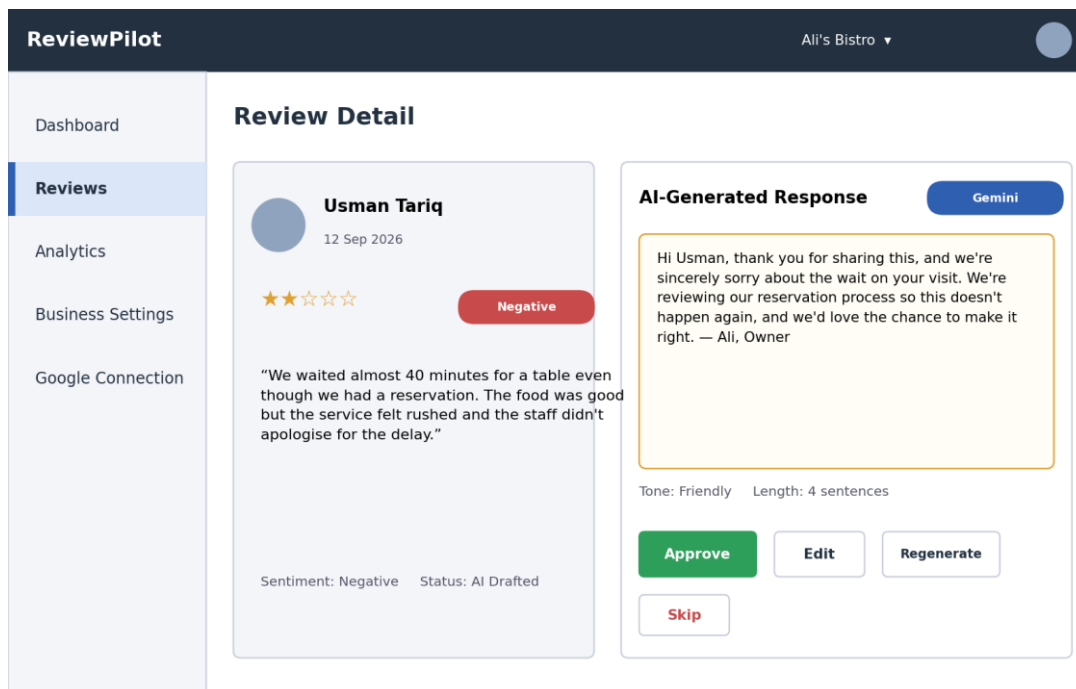


Figure 4 Review detail and AI response mockup

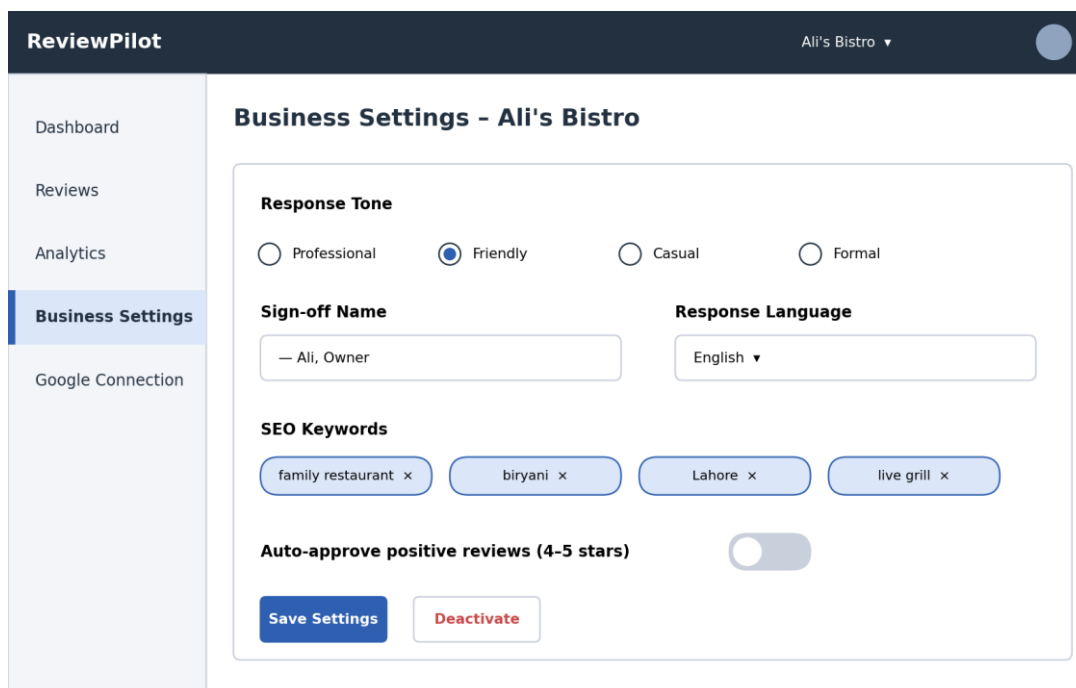


Figure 5 Business settings mockup

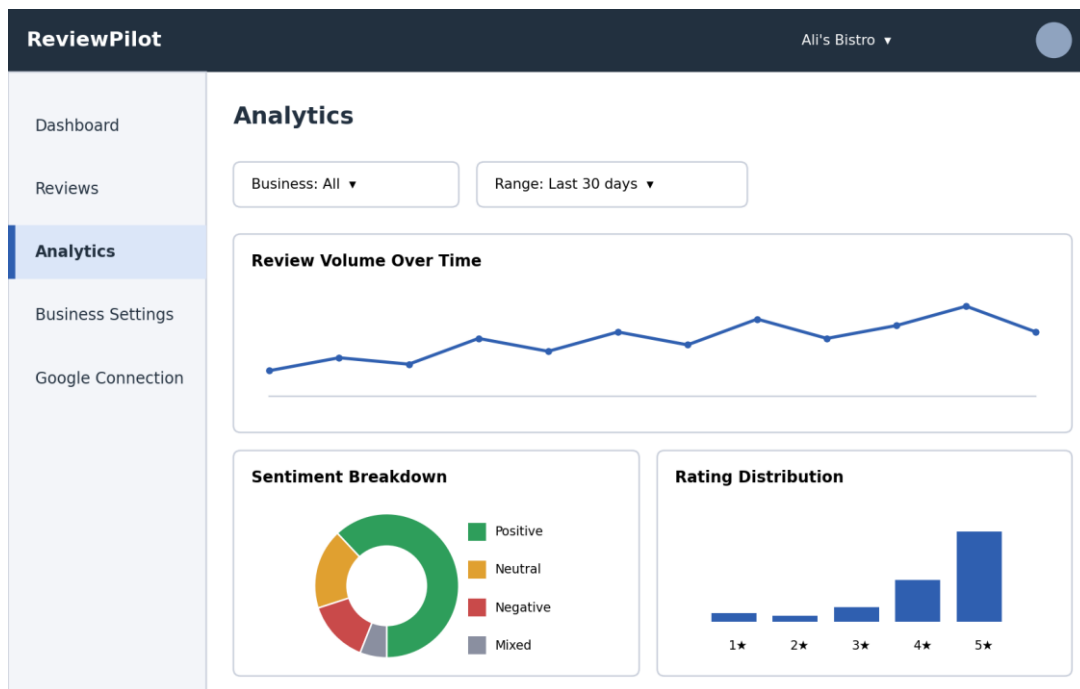


Figure 6 Analytics mockup

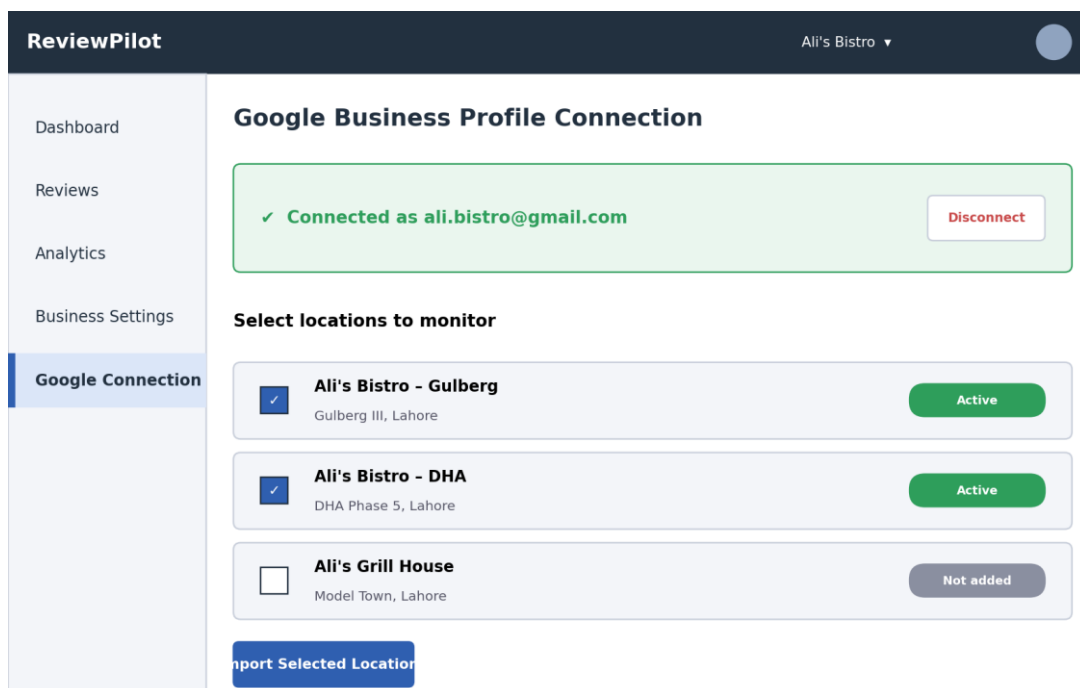


Figure 7 Google connection mockup

17. Conclusion

ReviewPilot aims to help small local businesses respond to Google reviews quickly, consistently, and professionally. By combining automatic review fetching with AI-drafted replies, it removes most of the effort while leaving the final decision with the owner. Its six modules cover authentication, Google integration, review fetching, AI response generation, response management, and analytics. The project is achievable with free-tier services and the MERN stack, and it gives the team practical experience with OAuth 2.0, large language models, and scheduled jobs. This scope document defines the boundaries of the first release and prepares the ground for the SRS and design phases.

18. References

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19. Plagiarism Report

The Turnitin plagiarism report of this scope document, obtained from the library staff, is to be attached on the following page.